

The Race between Man and Machine

Tasks, wages, and what a formal proof can establish

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<https://github.com/johnbarraza/ai-06-restrepo>

Road map for 20 minutes

- ① The aggregate task economy and equilibrium threshold
- ② Automation versus new tasks: opposite reallocations
- ③ The wage condition and a joint task extension with a visual check
- ④ Lean translation: checked theorems, proof steps, and limits
- ⑤ Hand verification and verdict

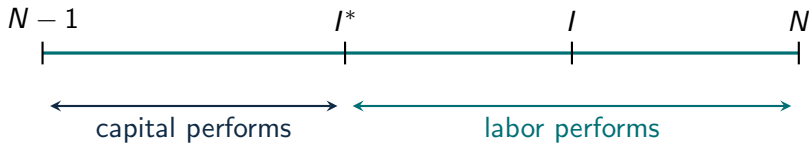
Question and unit of analysis

Question

Can automation raise output yet lower the wage? Can new tasks reverse the fall in labor's share?

Unlike the earlier weekly papers, the unit here is the **aggregate economy**: a continuum of production tasks and competitive factor markets.

The task space



Two frontiers

I is technological feasibility; I^* is actual capital use. They need not coincide.

From tasks to aggregate output

$$Y = \tilde{B} \left(\int_{N-1}^N y(i)^{(\sigma-1)/\sigma} di \right)^{\sigma/(\sigma-1)}.$$

- ▶ The task interval always has measure one.
- ▶ $\sigma > 0$ governs substitution across tasks; the $\sigma = 1$ case is understood by its limit.
- ▶ Automation changes *who performs* tasks; new task creation shifts the interval to higher i .

The resulting aggregate production function has capital and labor weights that depend on technology (Proposition 1).

The agents' problems

Competitive task producer For $i \leq I$,
compare factor costs

$$\min \left\{ R, \frac{W}{\gamma(i)} \right\}.$$

For $i > I$, only labor is feasible. Intermediate inputs also enter full unit cost.

Representative household Choose (C, L)
with $C = WL + RK$ to maximize the paper's
 $u(C, L)$. At an interior choice:

$$\nu'(L) = \frac{W}{C}.$$

Comparative advantage implies a threshold

$$\frac{W}{R} = \gamma(\tilde{I}), \quad I^* = \min\{I, \tilde{I}\}.$$

- ▶ $\gamma(i)$ strictly increases: labor has an advantage in more complex tasks.
- ▶ Below I^* : capital is cheaper; above I^* : labor is used.
- ▶ If $I < \tilde{I}$, technology constrains automation.
- ▶ If $\tilde{I} < I$, technology is slack; more feasible automation changes nothing locally.

Conditions behind the static results

- ① $\gamma(i)$ strictly increasing (Assumption 1).
- ② $\eta \rightarrow 0$ or $\zeta = 1$ (Assumption 2): homothetic factor demand.
- ③ $K < \bar{K}$ (Assumption 3): new tasks are used.

$\sigma > 0$, $\hat{\sigma} > 0$, $\varepsilon_L > 0$, and the household's convexity and regularity conditions also matter. Proposition 1 supplies a unique static equilibrium under these conditions.

Automation and reinstatement

Automation: $I \uparrow$

Capital takes over some existing labor tasks. In the constrained regime, W/R , labor share, and employment fall.

New tasks: $N \uparrow$

New complex tasks require labor and replace tasks with lower indices. W/R , labor share, employment, and the wage rise.

These are Propositions 2 and 3 static comparative statics, not unconditional predictions for a dynamic economy.

Where the wage equation comes from

$$(B9) \quad s_L d \ln W + (1 - s_L) d \ln R = d \ln Y|_{K,L},$$

$$(B10), dN = 0 : \quad d \ln W - d \ln R = -\frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} dl.$$

Let $x = d \ln W$, $z = d \ln R$, and $p = d \ln Y|_{K,L}$. Since $z = x - (x - z)$, (B9) becomes

$$x = p + (1 - s_L)(x - z).$$

Substituting (B10) gives the Proposition 3 wage decomposition. This algebra from two equations is the natural Lean target.

The wage trap: two forces

$$\underbrace{\frac{d \ln W}{dl}}_{\text{wage response}} = \underbrace{P_I}_{\text{productivity gain}} - \underbrace{\frac{(1 - s_L)\Lambda_I}{\hat{\sigma} + \varepsilon_L}}_{\text{displacement}}.$$

Exact local condition, $I^* = I < \tilde{I}$

The wage rises iff

$$P_I > \frac{(1 - s_L)\Lambda_I}{\hat{\sigma} + \varepsilon_L}.$$

Equality gives no local change; a reversed inequality gives a fall.

A different sign is fixed

$$\frac{d \ln(W/R)}{dl} = -\frac{\Lambda_l}{\hat{\sigma} + \varepsilon_L} < 0.$$

The labor share and employment move with $\omega = W/(RK)$ and therefore fall in the constrained case.

Interpretation

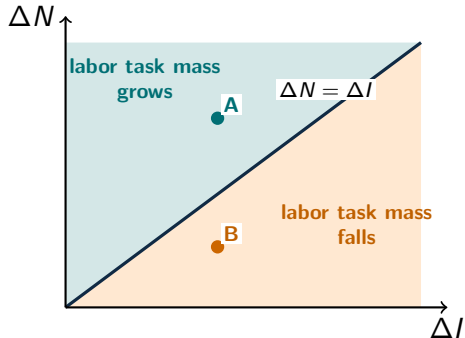
A falling labor share does not prove that the *level* of the real wage falls. Productivity and displacement compete in the wage equation.

Analytical and computational check

Illustrative case	P_I	displacement	$d \ln W / dI$
Large productivity gain	.400	.133	+.267
Small productivity gain	.050	.133	-.083
Equality	.133	.133	0

Same $s_L = .6$, $\Lambda_I = .5$, $\hat{\sigma} = 1$, $\varepsilon_L = .5$ in every row. Reproduce with python `analysis/wage_condition.py`. These inputs illustrate the sign identity; they are not solved equilibria.

Extension: automation and new tasks together



$$\Delta m_L = \Delta N - \Delta I.$$

Marked cases

A $\Delta I = .15, \Delta N = .25$
 $\Delta m_L = +.10$

B $\Delta I = .15, \Delta N = .08$
 $\Delta m_L = -.07$

Both frontiers must bind and lie inside their task intervals. The plot concerns task mass only; it does not determine wages or employment.

Where I did not trust the AI

Claim to test

“Automation necessarily reduces wages and labor’s share.”

- ▶ Labor share falls only when the automation frontier binds; at a slack frontier, a small increase in I has no effect.
- ▶ The wage can rise when the productivity gain dominates displacement.
- ▶ At $I^* = I = \tilde{I}$, use derivatives from each side.

Hand check: the decisive inequality

Restringo (Eqs 6)

Sea la subsección activa

$$I^* = \min\{I, \bar{I}\}$$

en la cual la tecnología innovadora y con N/R por

$$N-1 < I < \bar{I} < N \rightarrow I^* = I, \quad dN=0$$

Usando la ecuación (B9)

$$S_L = \frac{W}{W+R} \in (0,1) \quad A = \frac{\Lambda_I}{P+\varepsilon_L} > 0$$

lo cual es posible

Usando (B9) y (B10) con $dN=0$

$$S_L d \ln W + (1-S_L) d \ln R = d \ln Y / N_L \quad (B9)$$

$$d \ln W - d \ln R = -A d I \quad (B10) \Rightarrow \frac{d \ln W}{d I} = -A$$

Usando C.V. $Y = d \ln W$, $I = d \ln R$, $p = W/R$

$$I = x + A d I \quad (B10a)$$

$$S_L x + (1-S_L)(x + A d I) = p \quad (B9a)$$

$$\Rightarrow x = \frac{p}{1-(1-S_L)A} d I$$

$$\frac{d \ln W}{d I} = \frac{p}{1-(1-S_L)A} \frac{\Lambda_I}{P+\varepsilon_L}$$

lo que garantiza que sea positivo $\Rightarrow \frac{p}{1-(1-S_L)A} \frac{\Lambda_I}{P+\varepsilon_L} > 0$

En subsección pasiva donde W/R es un monopolio

entre N y I el valor real de la tecnología R_L es el efecto de desplazamiento

What equation (B10) fixes

$$\frac{d \ln(W/R)}{dI} = -\frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} < 0.$$

The wage falls relative to the rental price of capital.

My verdict after solving (B9)–(B10)

$$\frac{d \ln W}{dI} = P_I - D_I, \quad W \uparrow \iff P_I > D_I.$$

A fall in W/R does not imply a fall in W : productivity can dominate displacement.

Lean 1/6: what entered the checker

Pinned source

NBER Working Paper 22252, June 2017, 87 PDF pages. SHA-256 starts 441d0120.

- ▶ AppliedModelingLib generated the complete AR18RaceManMachine paper folder.
- ▶ Lean encodes the task cutoff, equation (13), factor income, and Appendix B's (B9)–(B10) bridge.
- ▶ The wage theorem assumes equation (13) and factor income as functional identities, with differentiability and nonzero conditions.

These are checked **support results**; a full equilibrium construction remains open.

Lean 2/6: equation $\rightarrow$ statement $\rightarrow$ proof

$$\underbrace{s_L d \ln W + (1 - s_L) d \ln R = p}_{(B9)}, \quad \underbrace{d \ln W - d \ln R = q}_{(B10)} \implies d \ln W = p + (1 - s_L)q.$$

```
theorem wage_identity_from_B9_B10
  (s dW dR productivity relativePrice : Real)
  (hB9 : s*dW + (1-s)*dR = productivity)
  (hB10 : dW-dR = relativePrice) :
  dW = productivity + (1-s)*relativePrice := by
  linear_combination hB9 + (1-s)*hB10
```

s encodes labor's share of net output; dW, dR encode the two log changes; p is productivity; q is the relative price change. The slide spells Lean's $\mathbb{R}$ notation as `Real`. Lean verifies the algebra *given* both premises.

Lean 3/6: from the model identities to (B9) and (B10)

- ▶ `StaticModel.lean`: $I^* = \min\{I, \tilde{I}\}$; when the frontier binds, raising I reduces the mass of labor tasks one for one.
- ▶ `B9_from_factorIncome`: differentiates $WL + RK = (1 - \eta)Y$ with fixed K, L .
- ▶ `B10_from_relativeDemandEquation`: differentiates equation (13), including the task integral, and uses $\omega = W/(RK)$.

The two bridges require explicit functional identities, derivatives, and nonzero denominators. The wage proof then combines (B9) and (B10) by `linear_combination`.

Lean 4/6: the exact wage condition

Set $q = -\Lambda_I/(\hat{\sigma} + \varepsilon_L)$ when the frontier binds. The checked specialization gives

$$d \ln W = P_I - \frac{(1 - s_L)\Lambda_I}{\hat{\sigma} + \varepsilon_L}.$$

Excerpt below: binders and repeated arguments omitted for legibility.

```
theorem automation_wage_rises_iff ... :  
  0 < dW <-> (1-s)*lambdaI/denominator  
    < productivity := by  
  rw [automation_wage_change ...]  
  constructor <;> intro h <;> linarith
```

The displayed lemma is conditional on (B9) and (B10). The stronger `automation_log_wage_rises_iff_from_equation13` derives them from explicit model identities. Positivity is needed for the economic sign interpretation.

Lean 5/6: results and open boundary

Machine results

`lake build +AR18RaceManMachine`: passed.

`paper_contribution.py check AR18RaceManMachine --fast`: passed.

The checked support theorems use only `propext`, `Classical.choice`, and `Quot.sound`.

What remains open

The full equilibrium, primitive derivation of equation (13), the dN wage term, and Proposition 3's capital threshold. No named result is claimed as fully proved; comparison of the source with Lean and final closeout remain open.

Lean 6/6: the joint task extension

$$m_1 - m_0 = \Delta N - \Delta I, \quad m_1 > m_0 \iff \Delta I < \Delta N.$$

Excerpt from JointTaskExtension.lean; the source file lists every feasibility hypothesis.

```
theorem joint_task_change_expands_labor_iff
  (N I costBefore costAfter newTasks automation : Real)
  ... :
  laborTaskMass N (equilibriumThreshold I costBefore) <
    laborTaskMass (N + newTasks)
    (equilibriumThreshold (I + automation) costAfter) <->
    automation < newTasks := by
  have h := (joint_task_change ...).1
  constructor <;> intro hsign <;> linarith
```

The slide spells Lean's $\mathbb{R}$ as Real. Lean checks the mass identity first. The final line turns it into a strict comparison. It assumes feasible task intervals and a binding technology frontier in both states. It proves no wage or employment result.

What the checks can and cannot conclude

- ▶ The computational script verifies arithmetic on the stated wage identity.
- ▶ Lean's compiler can verify a formal theorem only under its encoded assumptions and domain.
- ▶ Matching the NBER claim requires a separate comparison between the source and Lean statement.
- ▶ The full continuum equilibrium and all its comparative statics should be counted only if formally represented and proved.

Conclusion

Economic result

Automation reallocates tasks away from labor. With a binding frontier, labor share falls, while wages depend on the balance between productivity and displacement. New tasks reinstate labor.

Methodological result

The threshold, regime conditions, and sign inequality must all be explicit in any Lean translation or oral explanation.

Sources

Acemoglu, D., and P. Restrepo (2018). “The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment.” *AER* 108(6), 1488–1542.

Formalization source: NBER Working Paper 22252, revised June 2017, 87 pages. In particular Section 2, Propositions 1–3 and Appendix B.

Course requirement: <https://github.com/alexanderquispe/AI-Econ-Modeling/issues/5>